



POORNIMA
INSTITUTE OF ENGINEERING & TECHNOLOGY

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Basic Mechanical Engineering

251ME3-10/252ME3-10

Notes

Unit-4

Transmission of Power

Power transmission is a process to transmit motion from one shaft to another by using some connection between them like belt, rope, chain, and gears. To connect the shafts, mainly two types of connectors are used, one is flexible and other is rigid. In a flexible type of connection, there is a relative velocity between shaft and connectors due to slip and strain produced in the connectors. But in the case of rigid connection, there is no relative velocity between the connector and shaft.

Belt, rope, and chain are flexible connectors where gears are rigid connectors. Generally, belt, rope, and chain drives are used when the distance between the shafts are large and gears are used when the distance between the shafts is very small. The efficiency of a gear drive is much more.

The amount of power transmitted depends upon the following factors :

1. The velocity of the belt.
2. The tension under which the belt is placed on the pulleys.
3. The arc of contact between the belt and the smaller pulley.
4. The conditions under which the belt is used.

It may be noted that

- (a) The shafts should be properly in line to insure uniform tension across the belt section.
- (b) The pulleys should not be too close together, in order that the arc of contact on the smaller pulley may be as large as possible.
- (c) The pulleys should not be so far apart as to cause the belt to weigh heavily on the shafts, thus increasing the friction load on the bearings.
- (d) A long belt tends to swing from side to side, causing the belt to run out of the pulleys, which in turn develops crooked spots in the belt.
- (e) The tight side of the belt should be at the bottom, so that whatever sag is present on the loose side will increase the arc of contact at the pulleys.

(f) In order to obtain good results with flat belts, the maximum distance between the shafts should not exceed 10 metres and the minimum should not be less than 3.5 times the diameter of the larger pulley.

Selection of a Belt Drive

Following are the various important factors upon which the selection of a belt drive depends:

- 1.Speed of the driving and driven shafts,
- 2.Speed reduction ratio,
- 3.Power to be transmitted,
- 4.Centre distance between the shafts,
- 5.Positive drive requirements,
- 6.Shafts layout,
- 7.Space available, and
- 8.Service conditions.

Belt drive

Types of Belt Drives

- 1. Light drives.** These are used to transmit small powers at belt speeds upto about 10 m/s, as in agricultural machines and small machine tools.
- 2. Medium drives.** These are used to transmit medium power at belt speeds over 10 m/s but up to 22 m/s, as in machine tools.
- 3. Heavy drives.** These are used to transmit large powers at belt speeds above 22 m/s, as in compressors and generators.

In belt drive, the velocity of two shafts can be varied by variation of diameter of the pulley on which belt is mounted. If an un-stretched belt is mounted on the pulleys, the outer and inner faces of belt become in tension and compression,

Type of Belt Cross-sections

- (1) Flat belt
- (2) V- Belt

(3) Round belt

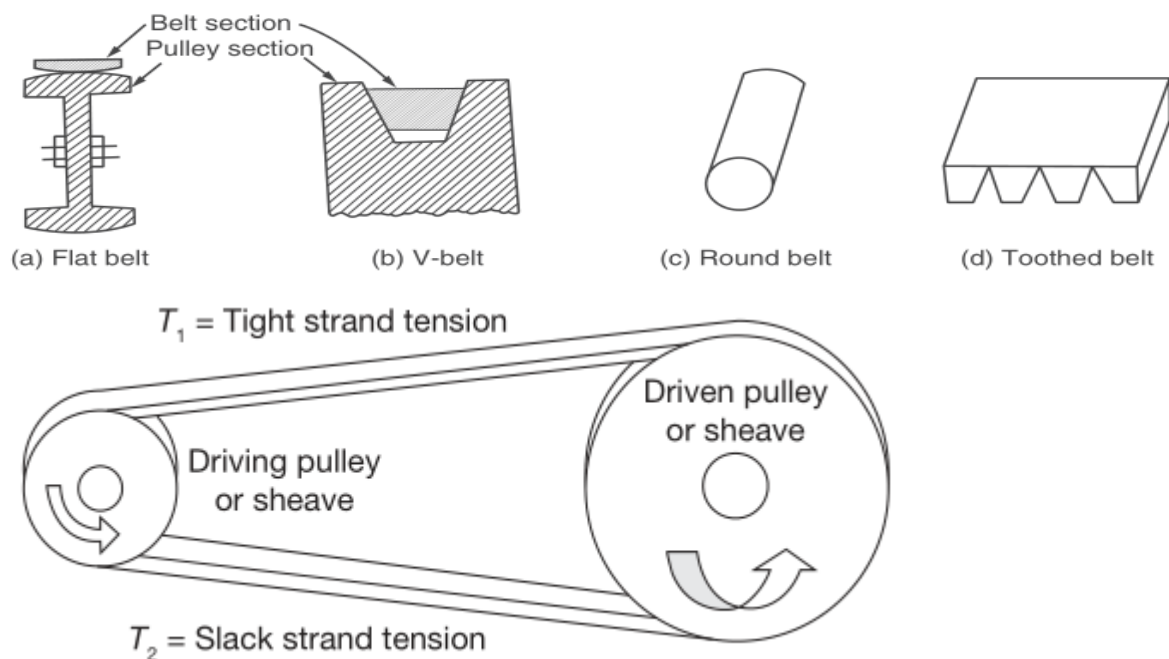
(4) Toothed belt

Flat belt drive, the rim of the pulley is slightly crowned which helps to keep the belt running centrally on the pulley rim.

V-belt drive, grooves are made on the rim of the pulley for wedging action. The belt does not touch the bottom of the groove. In multiple V-belt systems, more than one belt on the pulleys can be used to increase the power transmission capacity.

Round belts are generally made of rubber. This type of belt is generally used for light loads, such as in a sewing machine or a vacuum cleaner.

Timing belts or toothed belts use their teeth for power transmission, as opposed to friction. This configuration results in no slippage, and therefore, the driving and driven shafts remain synchronized



Material used for Belts:-

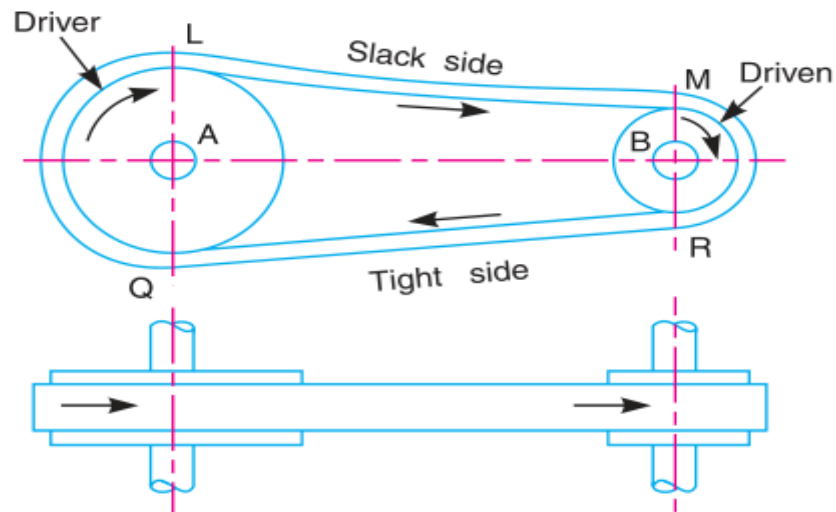
1. Leather belts.
2. Cotton or fabric belts
3. Rubber belt.
4. Balata belts.

Types of Flat Belt Drives

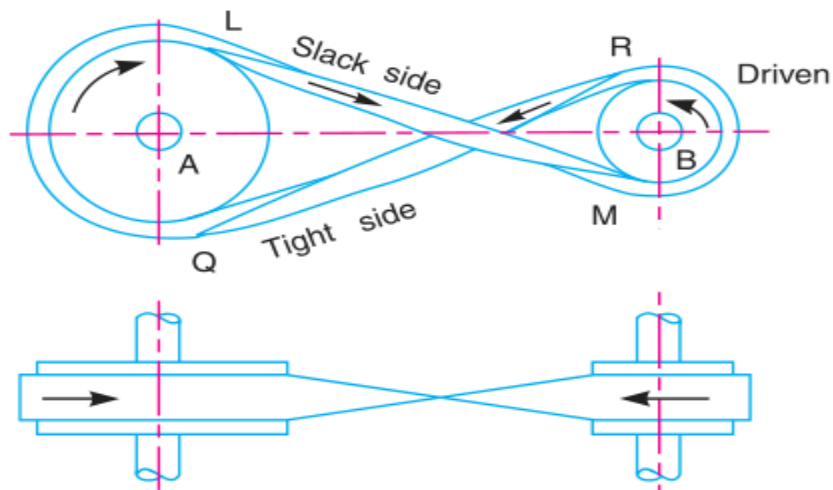
1. Open belt drive.

Flat Belt Drives

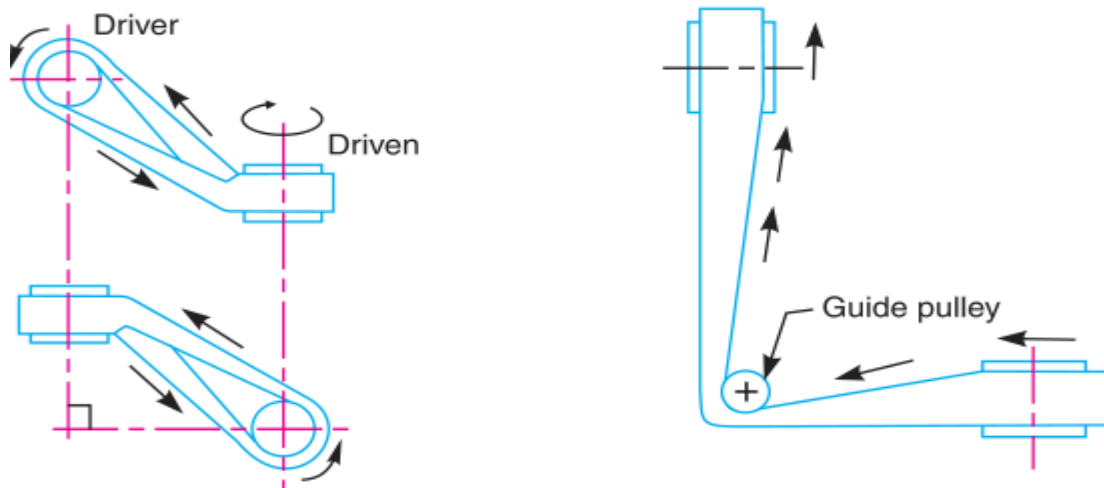
(a) Open belt drive



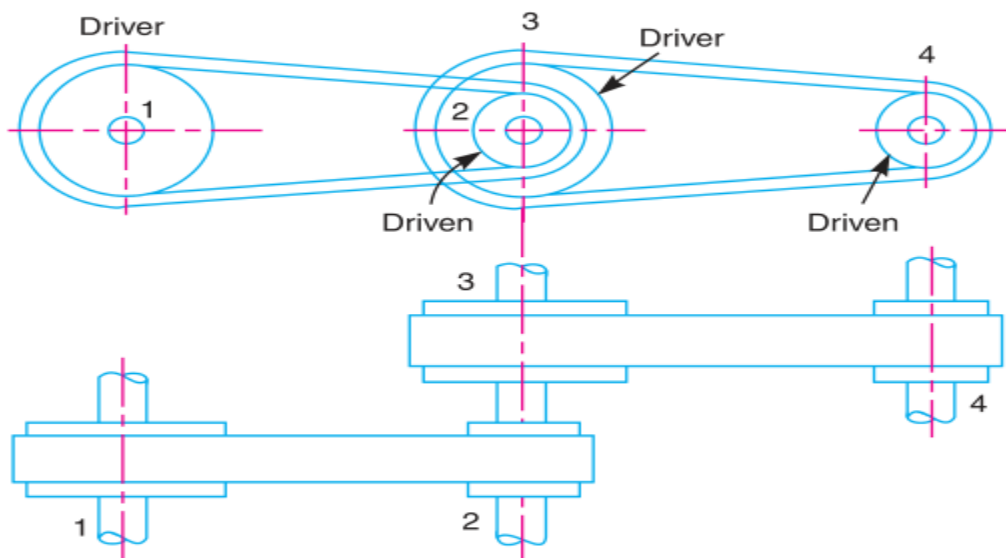
(b) Crossed belt drive



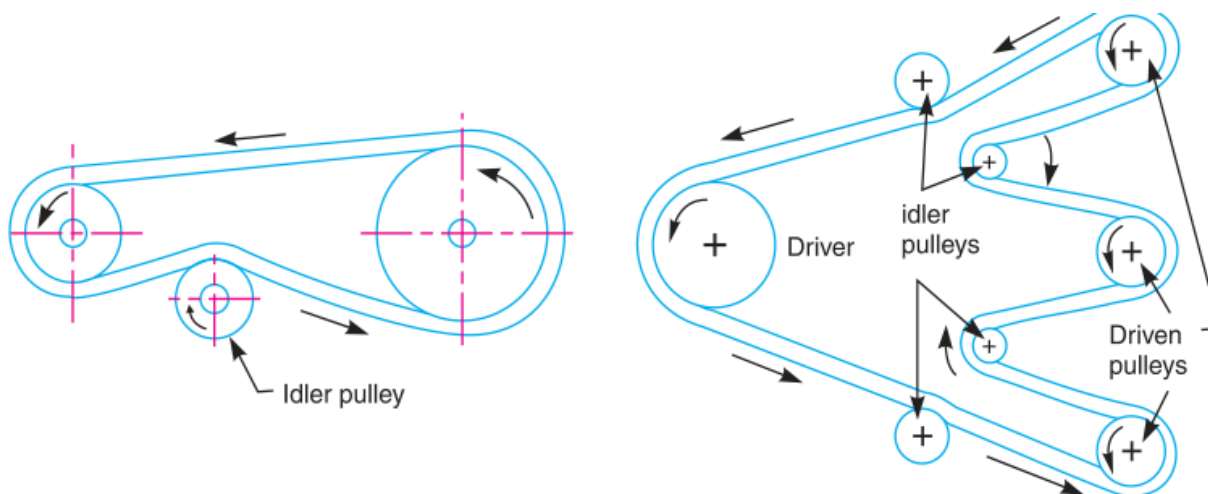
(c) Quarter turn belt drive



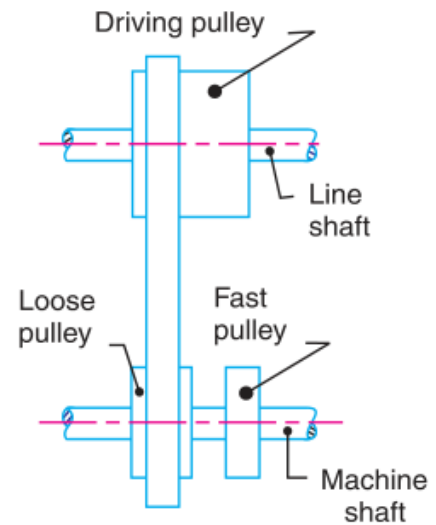
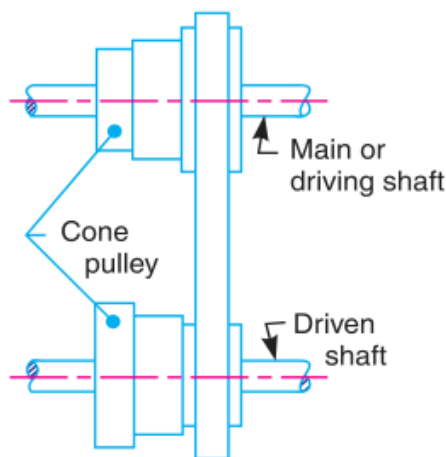
(d) Compound belt drive



5. Belt drive with idler pulleys



6. Stepped or cone pulley drive. 7. Fast and loose pulley drive



Velocity Ratio of Belt drive

Velocity ratio is the ratio of the speed of the driven pulley to that of the driving pulley.

Let N_1 is rotational speed of the driving pulley, N_2 is rotational speed of the driven pulley, D_1 is diameter of driving pulley, D_2 is diameter of driven pulley, t is the thickness of the belt.

$$V = \frac{\pi(D_1 + 2t)N_1}{60} = \frac{\pi(D_2 + 2t)N_2}{60}$$

$D_1N_1 = D_2N_2$; where t is very small in comparison to D , therefore it can be neglected.

$$V.R = \frac{N_2}{N_1} = \frac{D_1}{D_2}$$

Velocity ratio of compound Belt drive

Sometimes the power is transmitted from one shaft to another, through a number of pulleys as shown in Fig. 11.7. Consider a pulley 1 driving the pulley 2. Since the pulleys 2 and 3 are keyed to the same shaft, therefore the pulley 1 also drives the pulley 3 which, in turn, drives the pulley 4.

Let d_1 = Diameter of the pulley 1,
 N_1 = Speed of the pulley 1 in r.p.m.,
 d_2, d_3, d_4 , and N_2, N_3, N_4 = Corresponding values for pulleys 2, 3 and 4.
 We know that velocity ratio of pulleys 1 and 2,

$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \quad \dots(i)$$

Similarly, velocity ratio of pulleys 3 and 4,

$$\frac{N_4}{N_3} = \frac{d_3}{d_4} \quad \dots(ii)$$

Multiplying equations (i) and (ii),

$$\frac{N_2}{N_1} \times \frac{N_4}{N_3} = \frac{d_1}{d_2} \times \frac{d_3}{d_4}$$

$$\frac{N_4}{N_1} = \frac{d_1 \times d_3}{d_2 \times d_4} \quad \dots(\because N_2 = N_3, \text{ being keyed to the same shaft})$$

A little consideration will show, that if there are six pulleys, then

$$\frac{N_6}{N_1} = \frac{d_1 \times d_3 \times d_5}{d_2 \times d_4 \times d_6}$$

$$\frac{\text{Speed of last driven}}{\text{Speed of first driver}} = \frac{\text{Product of diameters of drivers}}{\text{Product of diameters of driven}}$$

Slip:

Some forward motion of the belt without carrying the driven pulley with it. This is called slip of the belt and is generally expressed as a percentage. The result of the belt slipping is to reduce the velocity ratio of the system or the effect of slip is a decrease in the speed of belt on driving shaft and the driven shaft.

Let

s_1 % = Slip between the driver and the belt, and

$s_2 \% =$ Slip between the belt and the follower.

$\therefore$ Velocity of the belt passing over the driver per second

$$v = \frac{\pi d_1 \cdot N_1}{60} - \frac{\pi d_1 \cdot N_1}{60} \times \frac{s_1}{100} = \frac{\pi d_1 \cdot N_1}{60} \left(1 - \frac{s_1}{100}\right) \quad \dots(i)$$

and velocity of the belt passing over the follower per second,

$$\frac{\pi d_2 \cdot N_2}{60} = v - v \times \frac{s_2}{100} = v \left(1 - \frac{s_2}{100}\right)$$

Substituting the value of v from equation (i),

$$\begin{aligned} \frac{\pi d_2 N_2}{60} &= \frac{\pi d_1 N_1}{60} \left(1 - \frac{s_1}{100}\right) \left(1 - \frac{s_2}{100}\right) \\ \frac{N_2}{N_1} &= \frac{d_1}{d_2} \left(1 - \frac{s_1}{100} - \frac{s_2}{100}\right) \quad \dots \left(\text{Neglecting } \frac{s_1 \times s_2}{100 \times 100} \right) \\ &= \frac{d_1}{d_2} \left(1 - \frac{s_1 + s_2}{100}\right) = \frac{d_1}{d_2} \left(1 - \frac{s}{100}\right) \end{aligned}$$

... (where $s = s_1 + s_2$, i.e. total percentage of slip)

If thickness of the belt (t) is considered, then

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t} \left(1 - \frac{s}{100}\right)$$

Creep

When belt passes from slack to tight side, a certain portion of belt extends and again contracts when belt passes through tight to slack side. Due to fluctuation in length of the belt, there is relative motion between belt and pulley surface. This relative motion is known as creep. Considering the creep, velocity ratio can be expressed as:

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}$$

Where N_1 and N_2 are the speeds of driving and driven pulleys, respectively; D_1 and D_2 are the diameters of the driver and driven pulleys, respectively; σ_1 and σ_2 are the stresses developed in the tight and slack side of the belt, respectively; and E is the modulus of elasticity of belt materials

Length of an Open Belt Drive

Open Belt Drive

The open belt drive is used to provide the same direction of rotation to the driven shaft as the direction of rotation of the driving shaft.

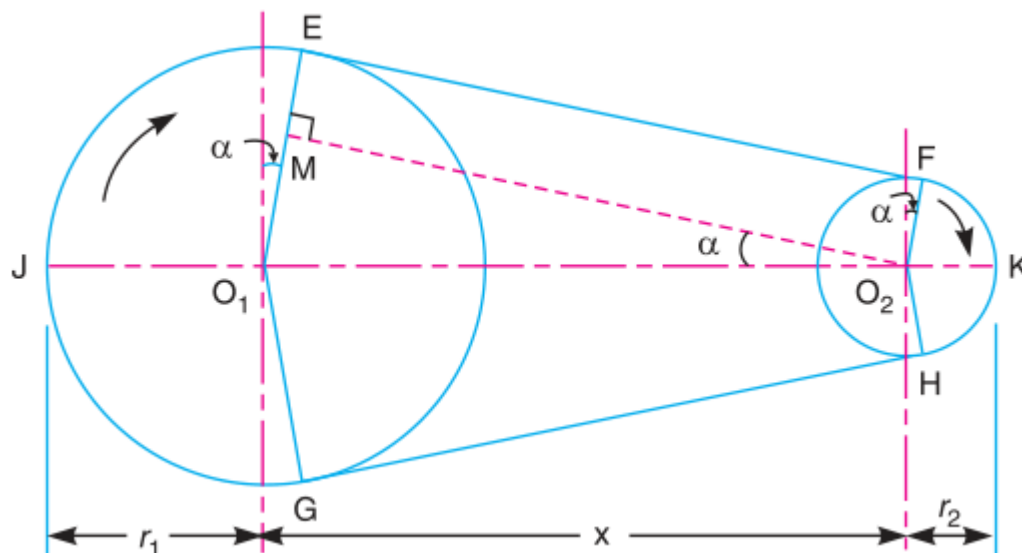
Let L is length of belt for open drive,

r is the radius of the smaller pulley,

R is radius of larger pulley,

C is center distance between pulleys,

β is angle subtended by each common tangent on the center of the pulley (CD or EF), AB is the line joining the centers of pulleys.



Let r_1 and r_2 = Radii of the larger and smaller pulleys,
 x = Distance between the centres of two pulleys (*i.e.* $O_1 O_2$), and
 L = Total length of the belt.

Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H as shown in figure. Through O_2 , draw $O_2 M$ parallel to FE .

From the geometry of the figure, we find that $O_2 M$ will be perpendicular to $O_1 E$.

Let the angle $MO_2 O_1 = \alpha$ radians.

We know that the length of the belt,

$$\begin{aligned} L &= \text{Arc } GJE + EF + \text{Arc } FKH + HG \\ &= 2 (\text{Arc } JE + EF + \text{Arc } FK) \end{aligned} \quad \dots(i)$$

From the geometry of the figure, we find that

$$\sin \alpha = \frac{O_1 M}{O_1 O_2} = \frac{O_1 E - EM}{O_1 O_2} = \frac{r_1 - r_2}{x}$$

Since α is very small, therefore putting

$$\sin \alpha = \alpha \text{ (in radians)} = \frac{r_1 - r_2}{x} \quad \dots(ii)$$

$$\therefore \text{Arc } JE = r_1 \left(\frac{\pi}{2} + \alpha \right) \quad \dots(iii)$$

$$\text{Similarly Arc } FK = r_2 \left(\frac{\pi}{2} - \alpha \right) \quad \dots(iv)$$

and

$$\begin{aligned} EF &= MO_2 = \sqrt{(O_1 O_2)^2 - (O_1 M)^2} = \sqrt{x^2 - (r_1 - r_2)^2} \\ &= x \sqrt{1 - \left(\frac{r_1 - r_2}{x} \right)^2} \end{aligned}$$

Expanding this equation by binomial theorem,

$$EF = x \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x} \right)^2 + \dots \right] = x - \frac{(r_1 - r_2)^2}{2x} \quad \dots(v)$$

Substituting the values of arc JE from equation (iii), arc FK from equation (iv) and EF from equation (v) in equation (i), we get

$$\begin{aligned} L &= 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + x - \frac{(r_1 - r_2)^2}{2x} + r_2 \left(\frac{\pi}{2} - \alpha \right) \right] \\ &= 2 \left[r_1 \times \frac{\pi}{2} + r_1 \cdot \alpha + x - \frac{(r_1 - r_2)^2}{2x} + r_2 \times \frac{\pi}{2} - r_2 \cdot \alpha \right] \\ &= 2 \left[\frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 - r_2) + x - \frac{(r_1 - r_2)^2}{2x} \right] \\ &= \pi (r_1 + r_2) + 2\alpha (r_1 - r_2) + 2x - \frac{(r_1 - r_2)^2}{x} \end{aligned}$$

Substituting the value of $\alpha = \frac{r_1 - r_2}{x}$ from equation (ii),

$$\begin{aligned} L &= \pi (r_1 + r_2) + 2 \times \frac{(r_1 - r_2)}{x} \times (r_1 - r_2) + 2x - \frac{(r_1 - r_2)^2}{x} \\ &= \pi (r_1 + r_2) + \frac{2(r_1 - r_2)^2}{x} + 2x - \frac{(r_1 - r_2)^2}{x} \\ &= \pi (r_1 + r_2) + 2x + \frac{(r_1 - r_2)^2}{x} \quad \dots(\text{In terms of pulley radii}) \\ &= \frac{\pi}{2} (d_1 + d_2) + 2x + \frac{(d_1 - d_2)^2}{4x} \quad \dots(\text{In terms of pulley diameters}) \end{aligned}$$

Length of a Cross Belt Drive

In a cross belt drive, both the pulleys rotate in opposite directions. Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H. Through O_2 , draw O_2M parallel to FE. From the geometry of the figure, we find that O_2M will be perpendicular to O_1E .

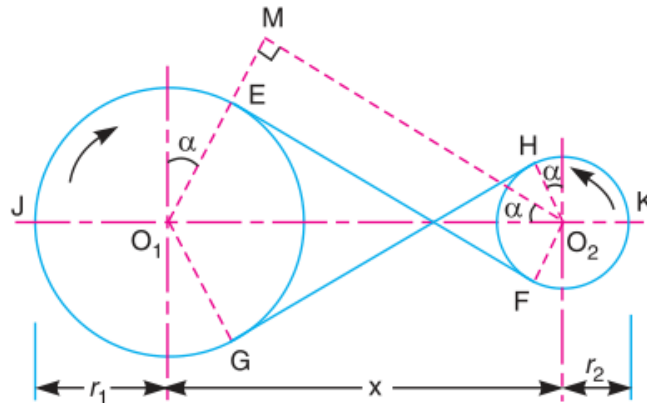


Fig. 11.12. Length of a cross belt drive.

Let r_1 and r_2 = Radii of the larger and smaller pulleys,
 x = Distance between the centres of two pulleys (i.e. $O_1 O_2$), and
 L = Total length of the belt.

Let the angle $MO_2 O_1 = \alpha$ radians.

We know that the length of the belt,

$$\begin{aligned} L &= \text{Arc } GJE + EF + \text{Arc } FKH + HG \\ &= 2 (\text{Arc } JE + EF + \text{Arc } FK) \end{aligned} \quad \dots(i)$$

From the geometry of the figure, we find that

$$\sin \alpha = \frac{O_1 M}{O_1 O_2} = \frac{O_1 E + EM}{O_1 O_2} = \frac{r_1 + r_2}{x}$$

Since α is very small, therefore putting

$$\sin \alpha = \alpha \text{ (in radians)} = \frac{r_1 + r_2}{x} \quad \dots(ii)$$

$$\therefore \text{Arc } JE = r_1 \left(\frac{\pi}{2} + \alpha \right) \quad \dots(iii)$$

$$\text{Similarly } \text{Arc } FK = r_2 \left(\frac{\pi}{2} + \alpha \right) \quad \dots(iv)$$

$$\begin{aligned} EF &= MO_2 = \sqrt{(O_1 O_2)^2 - (O_1 M)^2} = \sqrt{x^2 - (r_1 + r_2)^2} \\ &= x \sqrt{1 - \left(\frac{r_1 + r_2}{x} \right)^2} \end{aligned}$$

Expanding this equation by binomial theorem,

$$EF = x \left[1 - \frac{1}{2} \left(\frac{r_1 + r_2}{x} \right)^2 + \dots \right] = x - \frac{(r_1 + r_2)^2}{2x} \quad \dots(v)$$

Substituting the values of arc JE from equation (iii), arc FK from equation (iv) and EF from equation (v) in equation (i), we get

$$\begin{aligned} L &= 2 \left[r_1 \left(\frac{\pi}{2} + \alpha \right) + x - \frac{(r_1 + r_2)^2}{2x} + r_2 \left(\frac{\pi}{2} + \alpha \right) \right] \\ &= 2 \left[r_1 \times \frac{\pi}{2} + r_1 \cdot \alpha + x - \frac{(r_1 + r_2)^2}{2x} + r_2 \times \frac{\pi}{2} + r_2 \cdot \alpha \right] \\ &= 2 \left[\frac{\pi}{2} (r_1 + r_2) + \alpha (r_1 + r_2) + x - \frac{(r_1 + r_2)^2}{2x} \right] \\ &= \pi (r_1 + r_2) + 2\alpha (r_1 + r_2) + 2x - \frac{(r_1 + r_2)^2}{x} \end{aligned}$$

Substituting the value of $\alpha = \frac{r_1 + r_2}{x}$ from equation (ii),

$$\begin{aligned} L &= \pi (r_1 + r_2) + \frac{2(r_1 + r_2)}{x} \times (r_1 + r_2) + 2x - \frac{(r_1 + r_2)^2}{x} \\ &= \pi (r_1 + r_2) + \frac{2(r_1 + r_2)^2}{x} + 2x - \frac{(r_1 + r_2)^2}{x} \\ &= \pi (r_1 + r_2) + 2x + \frac{(r_1 + r_2)^2}{x} \quad \dots(\text{In terms of pulley radii}) \\ &= \frac{\pi}{2} (d_1 + d_2) + 2x + \frac{(d_1 + d_2)^2}{4x} \quad \dots(\text{In terms of pulley diameters}) \end{aligned}$$

Power Transmitted by a Belt

The driving pulley (or driver) A and the driven pulley (or follower) B. We have already discussed that the driving pulley pulls the belt from one side and delivers the same to the other side. It is thus obvious that the tension on the former side (i.e. tight side) will be greater than the latter side (i.e. slack side). Let T_1 and T_2 = Tensions in the tight and slack side of the belt respectively in Newtons

r_1 and r_2 = Radii of the driver and follower respectively, and
 v = Velocity of the belt in m/s.

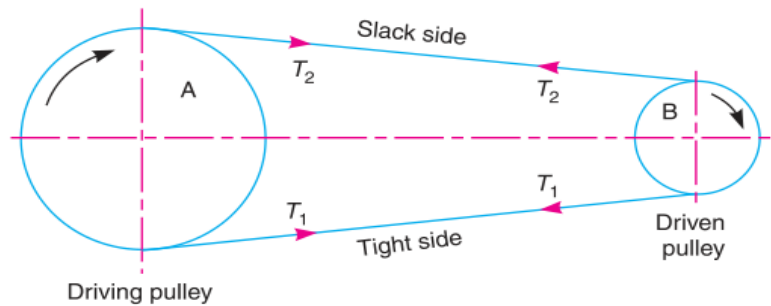


Fig. 11.14. Power transmitted by a belt.

The effective turning (driving) force at the circumference of the follower is the difference between the two tensions (*i.e.* $T_1 - T_2$).

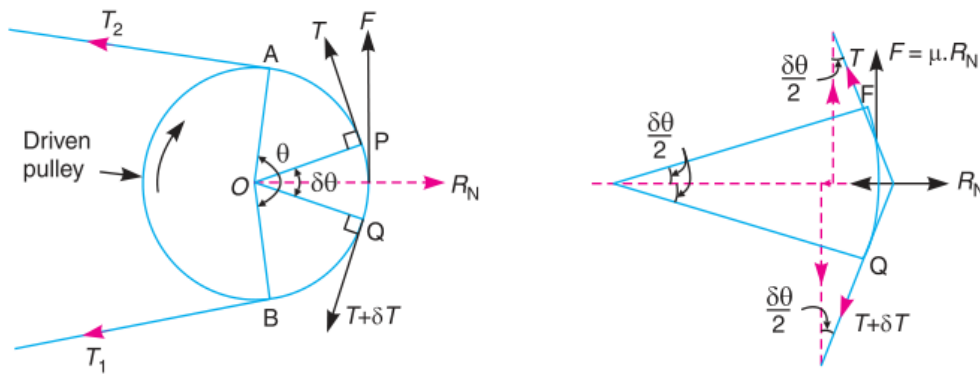
$$\therefore \text{Work done per second} = (T_1 - T_2) v \text{ N-m/s}$$

and power transmitted, $P = (T_1 - T_2) v \text{ W}$...($\because 1 \text{ N-m/s} = 1 \text{ W}$)

A little consideration will show that the torque exerted on the driving pulley is $(T_1 - T_2) r_1$. Similarly, the torque exerted on the driven pulley *i.e.* follower is $(T_1 - T_2) r_2$.

Ratio of Driving Tensions For Flat Belt Drive

Consider a driven pulley rotating in the clockwise direction as shown



Ratio of driving tensions for flat belt.

Let

T_1 = Tension in the belt on the tight side,

T_2 = Tension in the belt on the slack side, and

θ = Angle of contact in radians (*i.e.* angle subtended by the arc AB, along which the belt touches the pulley at the centre).

Now consider a small portion of the belt PQ, subtending an angle $\delta\theta$ at the centre of the pulley as shown. The belt PQ is in equilibrium under the following forces :

1. Tension T in the belt at P,
2. Tension $(T + \delta T)$ in the belt at Q,
3. Normal reaction R_N , and
4. Frictional force, $F = \mu \times R_N$, where μ is the coefficient of friction between the belt and pulley.

Resolving all the forces horizontally and equating the same,

$$R_N = (T + \delta T) \sin \frac{\delta\theta}{2} + T \sin \frac{\delta\theta}{2} \quad \dots(i)$$

Since the angle $\delta\theta$ is very small, therefore putting $\sin \delta\theta / 2 = \delta\theta / 2$ in equation (i),

$$R_N = (T + \delta T) \frac{\delta\theta}{2} + T \times \frac{\delta\theta}{2} = \frac{T \cdot \delta\theta}{2} + \frac{\delta T \cdot \delta\theta}{2} + \frac{T \cdot \delta\theta}{2} = T \cdot \delta\theta \quad \dots(ii)$$

$\dots \left(\text{Neglecting } \frac{\delta T \cdot \delta\theta}{2} \right)$

Now resolving the forces vertically, we have

$$\mu \times R_N = (T + \delta T) \cos \frac{\delta\theta}{2} - T \cos \frac{\delta\theta}{2} \quad \dots(iii)$$

Since the angle $\delta\theta$ is very small, therefore putting $\cos \delta\theta / 2 = 1$ in equation (iii),

$$\mu \times R_N = T + \delta T - T = \delta T \quad \text{or} \quad R_N = \frac{\delta T}{\mu} \quad \dots(iv)$$

Equating the values of R_N from equations (ii) and (iv),

$$T \cdot \delta\theta = \frac{\delta T}{\mu} \quad \text{or} \quad \frac{\delta T}{T} = \mu \cdot \delta\theta$$

Integrating both sides between the limits T_2 and T_1 and from 0 to θ respectively,

Integrating both sides between the limits T_2 and T_1 and from 0 to θ respectively,

$$i.e. \quad \int_{T_2}^{T_1} \frac{\delta T}{T} = \mu \int_0^\theta \delta\theta \quad \text{or} \quad \log_e \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta \quad \text{or} \quad \frac{T_1}{T_2} = e^{\mu \cdot \theta} \quad \dots(v)$$

Equation (v) can be expressed in terms of corresponding logarithm to the base 10, i.e.

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta$$

Centrifugal Tension

Since the belt continuously runs over the pulleys,

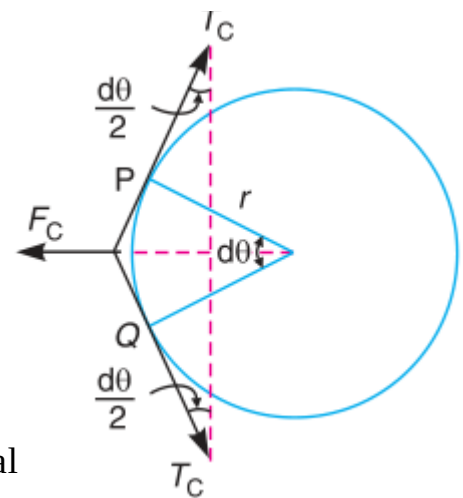
therefore, some centrifugal force is caused,

whose effect is to increase the tension on both,

tight as well as the slack sides. The tension caused

by centrifugal force is called centrifugal tension.

At lower belt speeds (less than 10 m/s), the centrifugal



tension is very small, but at higher belt speeds (more than 10 m/s), its effect is considerable and thus should be taken into account. Consider a small portion PQ of the belt subtending an angle $d\theta$ the centre of the pulley

Let m = Mass of the belt per unit length in kg,
 v = Linear velocity of the belt in m/s,
 r = Radius of the pulley over which the belt runs in metres, and
 T_C = Centrifugal tension acting tangentially at P and Q in newtons.

We know that length of the belt PQ

$$= r \cdot d\theta$$

and mass of the belt PQ $= m \cdot r \cdot d\theta$

$\therefore$ Centrifugal force acting on the belt PQ ,

$$F_C = (m \cdot r \cdot d\theta) \frac{v^2}{r} = m \cdot d\theta \cdot v^2$$

The centrifugal tension T_C acting tangentially at P and Q keeps the belt in equilibrium.

Now resolving the forces (*i.e.* centrifugal force and centrifugal tension) horizontally and equating the same, we have

$$T_C \sin\left(\frac{d\theta}{2}\right) + T_C \sin\left(\frac{d\theta}{2}\right) = F_C = m \cdot d\theta \cdot v^2$$

Since the angle $d\theta$ is very small, therefore, putting $\sin\left(\frac{d\theta}{2}\right) = \frac{d\theta}{2}$, in the above expression,

$$2T_C \left(\frac{d\theta}{2}\right) = m \cdot d\theta \cdot v^2 \quad \text{or} \quad T_C = m \cdot v^2$$

Notes : 1. When the centrifugal tension is taken into account, then total tension in the tight side,

$$T_{t1} = T_1 + T_C$$

and total tension in the slack side,

$$T_{t2} = T_2 + T_C$$

2. Power transmitted, $P = (T_{t1} - T_{t2}) v$... (in watts)
 $= [(T_1 + T_C) - (T_2 + T_C)] v = (T_1 - T_2) v$... (same as before)

Thus we see that centrifugal tension has no effect on the power transmitted.

3. The ratio of driving tensions may also be written as

$$2.3 \log \left(\frac{T_{t1} - T_C}{T_{t2} - T_C} \right) = \mu \cdot \theta$$

where

T_{t1} = Maximum or total tension in the belt.

Maximum Tension in the Belt

A little consideration will show that the maximum tension in the belt (T) is equal to the total tension in the tight side of the belt (T_{t1}).

Let σ = Maximum safe stress in N/mm²,
 b = Width of the belt in mm, and
 t = Thickness of the belt in mm.

We know that maximum tension in the belt,

$$T = \text{Maximum stress} \times \text{cross-sectional area of belt} = \sigma \cdot b \cdot t$$

When centrifugal tension is neglected, then

$$T \text{ (or } T_{t1}) = T_1, \text{ i.e. Tension in the tight side of the belt}$$

and when centrifugal tension is considered, then

$$T \text{ (or } T_{t1}) = T_1 + T_C$$

Condition For the Transmission of Maximum Power

We know that power transmitted by a belt,

$$P = (T_1 - T_2) v \quad \dots(i)$$

where

T_1 = Tension in the tight side of the belt in newtons,

T_2 = Tension in the slack side of the belt in newtons, and

v = Velocity of the belt in m/s.

From Art. 11.14, we have also seen that the ratio of driving tensions is

$$\frac{T_1}{T_2} = e^{\mu \cdot \theta} \quad \text{or} \quad T_2 = \frac{T_1}{e^{\mu \cdot \theta}} \quad \dots(ii)$$

Substituting the value of T_2 in equation (i),

$$P = \left(T_1 - \frac{T_1}{e^{\mu \cdot \theta}} \right) v = T_1 \left(1 - \frac{1}{e^{\mu \cdot \theta}} \right) v = T_1 \cdot v \cdot C \quad \dots(iii)$$

where

$$C = 1 - \frac{1}{e^{\mu \cdot \theta}}$$

We know that

$$T_1 = T - T_C$$

where

T = Maximum tension to which the belt can be subjected in newtons, and

T_C = Centrifugal tension in newtons.

Substituting the value of T_1 in equation (iii),

$$\begin{aligned} P &= (T - T_C) v \cdot C \\ &= (T - m \cdot v^2) v \cdot C = (T \cdot v - m v^3) C \quad \dots \text{(Substituting } T_C = m \cdot v^2) \end{aligned}$$

For maximum power, differentiate the above expression with respect to v and equate to zero, i.e.

$$\frac{dP}{dv} = 0 \quad \text{or} \quad \frac{d}{dv} (T \cdot v - m v^3) C = 0$$

$$\therefore T - 3 m \cdot v^2 = 0$$

or

$$T - 3 T_C = 0 \quad \text{or} \quad T = 3 T_C \quad \dots(iv)$$

It shows that when the power transmitted is maximum, 1/3rd of the maximum tension is absorbed as centrifugal tension.

Notes : 1. We know that $T_1 = T - T_C$ and for maximum power, $T_C = \frac{T}{3}$.

$\therefore$

$$T_1 = T - \frac{T}{3} = \frac{2T}{3}$$

2. From equation (iv), the velocity of the belt for the maximum power,

$$v = \sqrt{\frac{T}{3m}}$$

Initial Tension in the Belt

When a belt is wound round the two pulleys (i.e. driver and follower), its two ends are joined together ; so that the belt may continuously move over the pulleys, since the motion of the belt from the driver and the follower is governed by a firm grip, due to friction between the belt and the pulleys. In order to increase this grip, the belt is tightened up. At this stage, even when the pulleys are stationary, the belt is subjected to some tension, called initial tension.

When the driver starts rotating, it pulls the belt from one side (increasing tension in the belt on this side) and delivers it to the other side (decreasing the tension in the belt on that side). The increased tension in one side of the belt is called tension in tight side and the decreased tension in the other side of the belt is called tension in the slack side.

Let T_0 = Initial tension in the belt,
 T_1 = Tension in the tight side of the belt,
 T_2 = Tension in the slack side of the belt, and
 α = Coefficient of increase of the belt length per unit force.

A little consideration will show that the increase of tension in the tight side

$$= T_1 - T_0$$

and increase in the length of the belt on the tight side

$$= \alpha (T_1 - T_0) \quad \dots(i)$$

Similarly, decrease in tension in the slack side

$$= T_0 - T_2$$

and decrease in the length of the belt on the slack side

$$= \alpha (T_0 - T_2) \quad \dots(ii)$$

Assuming that the belt material is perfectly elastic such that the length of the belt remains constant, when it is at rest or in motion, therefore increase in length on the tight side is equal to decrease in the length on the slack side. Thus, equating equations (i) and (ii),

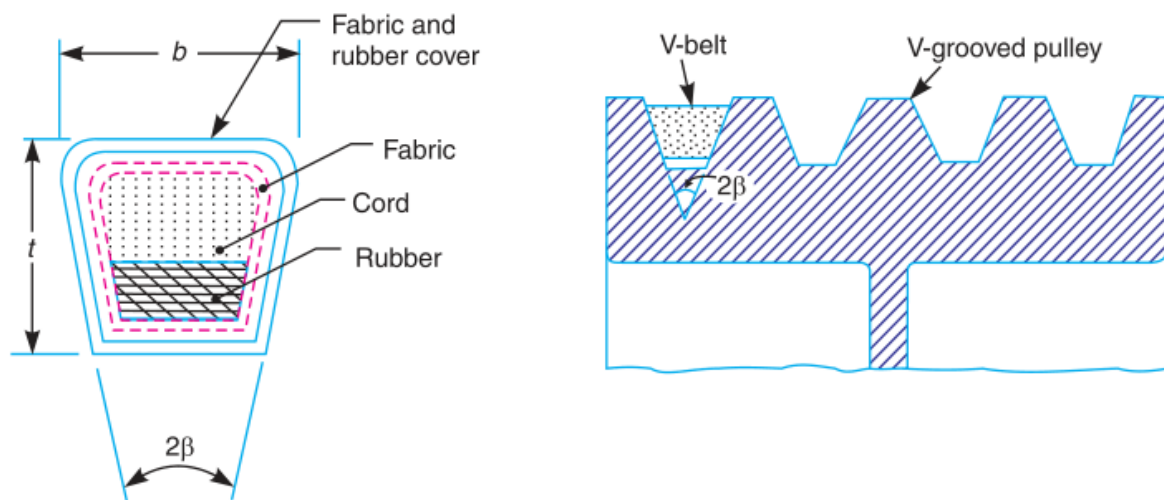
$$\alpha (T_1 - T_0) = \alpha (T_0 - T_2) \quad \text{or} \quad T_1 - T_0 = T_0 - T_2$$

$$\therefore T_0 = \frac{T_1 + T_2}{2} \quad \dots(\text{Neglecting centrifugal tension})$$

$$= \frac{T_1 + T_2 + 2T_C}{2} \quad \dots(\text{Considering centrifugal tension})$$

V-belt drive

V-belt is mostly used in factories and workshops where a great amount of power is to be transmitted from one pulley to another when the two pulleys are very near to each other. The V-belts are made of fabric and cords moulded in rubber and covered with fabric and rubber. These belts are moulded to a trapezoidal shape and are made endless. These are particularly suitable for short drives i.e. when the shafts are at a short distance apart. The included angle for the V-belt is usually from $30^\circ - 40^\circ$. In case of flat belt drive, the belt runs over the pulleys whereas in case of V-belt drive, the rim of the pulley is grooved in which the V-belt runs. The effect of the groove is to increase the frictional grip of the V-belt on the pulley and thus to reduce the tendency of slipping.



Advantages and Disadvantages of V-belt Drive Over Flat Belt Drive

Advantages

1. The V-belt drive gives compactness due to the small distance between the centres of pulleys.
2. The drive is positive, because the slip between the belt and the pulley groove is negligible.
3. Since the V-belts are made endless and there is no joint trouble, therefore the drive is smooth.

4. It provides longer life, 3 to 5 years.
5. It can be easily installed and removed.
6. The operation of the belt and pulley is quiet.
7. The belts have the ability to cushion the shock when machines are started.
8. The high velocity ratio (maximum 10) may be obtained.
9. The wedging action of the belt in the groove gives high value of limiting ratio of tensions. Therefore the power transmitted by V-belts is more than flat belts for the same coefficient of friction, arc of contact and allowable tension in the belts.
10. The V-belt may be operated in either direction with tight side of the belt at the top or bottom. The centre line may be horizontal, vertical or inclined.

Disadvantages

1. The V-belt drive cannot be used with large centre distances.
2. The V-belts are not so durable as flat belts.
3. The construction of pulleys for V-belts is more complicated than pulleys for flat belts.
4. Since the V-belts are subjected to certain amount of creep, therefore these are not suitable for constant speed application such as synchronous machines, and timing devices.
5. The belt life is greatly influenced with temperature changes, improper belt tension and mismatching of belt lengths.
6. The centrifugal tension prevents the use of V-belts at speeds below 5 m/s and above 50 m/s

Ratio of Driving Tensions for V-belt

A V-belt with a grooved pulley is shown in Fig. 11.20.

Let R_1 = Normal reaction between the belt and sides of the groove.

R = Total reaction in the plane of the groove.

2β = Angle of the groove.

μ = Coefficient of friction between the belt and sides of the groove.

Resolving the reactions vertically to the groove,

$$R = R_1 \sin \beta + R_1 \sin \beta = 2 R_1 \sin \beta$$

or
$$R_1 = \frac{R}{2 \sin \beta}$$

We know that the frictional force

$$= 2\mu \cdot R_1 = 2\mu \times \frac{R}{2 \sin \beta} = \frac{\mu \cdot R}{\sin \beta} = \mu \cdot R \operatorname{cosec} \beta$$

Consider a small portion of the belt, as in Art. 11.14, subtending an angle $\delta\theta$ at the centre. The tension on one side will be T and on the other side $T + \delta T$. Now proceeding as in Art. 11.14, we get the frictional resistance equal to $\mu \cdot R \operatorname{cosec} \beta$ instead of $\mu \cdot R$. Thus the relation between T_1 and T_2 for the V-belt drive will be

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta \operatorname{cosec} \beta$$

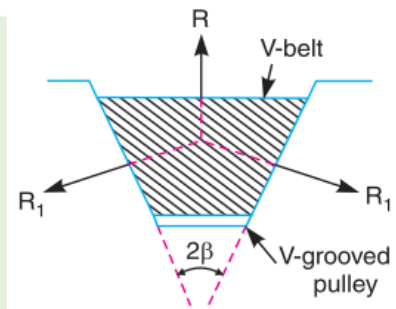


Fig. 11.20.

Rope drive

The rope drives are widely used to transmit large amount of power from one pulley to another, over a long distance. Ropes are made of cotton, hemp, manila or steel wire. The frictional grip in the case of rope drive is more than the belt drive. One of the main advantages of rope drive is that a number of drives can be taken from one driving pulley. For example, in many spinning mills, the line shaft on each floor is driven by ropes passing directly from the main engine pulley on the ground floor. For a rope drive, the groove angle of the pulley is normally 45° .

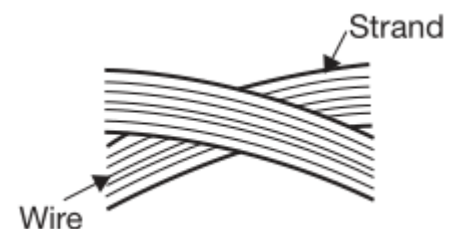
Rope drive is very similar to belt drive. It is classified as

(i) Fiber ropes

(ii) Wire ropes.

The fibre ropes operate successfully when the pulleys

are about 60 metres apart, while the wire ropes are used when the pulleys are upto 150 metres apart.



Fibre Ropes

The ropes for transmitting power are usually made from fibrous materials such as hemp, manila and cotton. Since the hemp and manila fibres are rough, therefore the ropes made from these fibres are not very flexible and possess poor mechanical properties. The hemp ropes have less strength as compared to manila ropes. When the hemp and manila ropes are bent over the sheave (or pulley), there is some sliding of fibres, causing the rope to wear and chafe internally. In order to minimise this defect, the rope fibres are lubricated with a tar, tallow or graphite. The lubrication also makes the rope moisture proof. The hemp ropes are suitable only for hand operated hoisting machinery and as tie ropes for lifting tackle, hooks etc. The cotton ropes are very soft and smooth. The lubrication of cotton ropes is not necessary. But if it is done, it reduces the external wear between the rope and the grooves of its sheaves. It may be noted that manila ropes are more durable and stronger than cotton ropes. The cotton ropes are costlier than manila ropes.

Advantages of Fibre Rope Drives

1. They give smooth, steady and quiet service.
2. They are little affected by outdoor conditions.
3. The shafts may be out of strict alignment.
4. The power may be taken off in any direction and in fractional parts of the whole amount.
5. They give high mechanical efficiency.

Steel wire Ropes

When a large amount of power is to be transmitted over long distances from one pulley to another (i.e. when the pulleys are upto 150 metres apart), then wire ropes are used. The wire ropes are widely used in elevators, mine hoists, cranes,

conveyors, hauling devices and suspension bridges. The wire ropes run on grooved pulleys but they rest on the bottom of the *grooves and are not wedged between the sides of the grooves. A group of wires makes a strand and strands make a rope. Each strand is twisted with other strands. The rope may have 3-strands or 9-strands, and each strand may have 7-19 wires, depends on its application. The wire ropes have the following advantage over cotton ropes.

1. These are lighter in weight,
2. These offer silent operation,
3. These can withstand shockloads,
4. These are more reliable,
5. They do not fail suddenly,
6. These are more durable,
7. The efficiency is high, and 8. The cost is low.

Ratio of Driving Tensions for Rope Drive

The ratio of driving tensions for the rope drive may be obtained in the similar way as V-belts. that the ratio of driving tensions is, where, μ , θ and β have usual meanings.

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta \operatorname{cosec} \beta$$

Difference between Rope drive and Belt drive

	Rope drive	Belt drive
1.	More suitable for long center distances between the shafts (more than 15m)	Suitable for shorter center distances (not more than 15 m)
2.	Frictional grip is more	Frictional grip is less
3.	Occurrence of slip is not possible	Occurrence of slip is possible
4.	More power is transmitted	Moderate power is transmitted
5.	Number of separate drives may be taken from one driving pulley	It is not possible to take number of separate drives from one driving pulley
6.	Operating speed is less than 3 m/sec	Operating speed is up to 24 m/sec

Difference between V-belt and Flat belt

	V-belt	Flat belt
1.	Suitable for shorter center distances between the shafts	Suitable for comparatively longer center distances
2.	Trapezoidal section	Rectangular section
3.	Frictional grip is more	Frictional grip is less
4.	Since more than one belt can be used, power transmitted will be more	Since single belt can be used power transmitted is less
5.	Velocity ratio is high	Velocity ratio is low
6.	Occurrence of slip is seldom possible	Slip occurs easily

Q.1 An engine, running at 150 r.p.m., drives a line shaft by means of a belt. The engine pulley is 750 mm diameter and the pulley on the line shaft being 450 mm. A 900 mm diameter pulley on the line shaft drives a 150 mm diameter pulley keyed to a dynamo shaft. Find the speed of the dynamo shaft, when 1. there is no slip, and 2. there is a slip of 2% at each drive.

Solution. Given : $N_1 = 150$ r.p.m. ; $d_1 = 750$ mm ; $d_2 = 450$ mm ; $d_3 = 900$ mm ; $d_4 = 150$ mm

The arrangement of belt drive is shown in Fig. 11.10.

Let N_4 = Speed of the dynamo shaft .

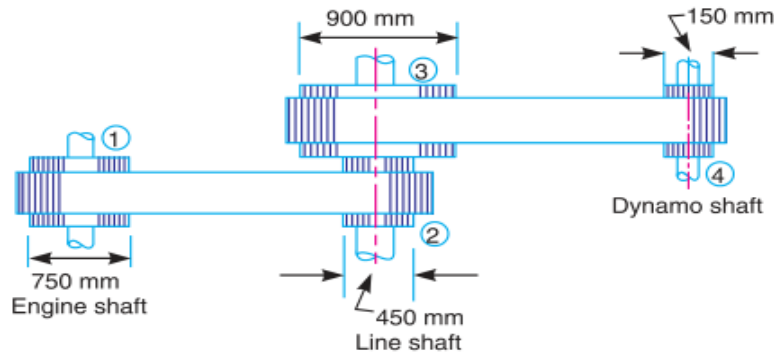


Fig. 11.10

1. When there is no slip

We know that
$$\frac{N_4}{N_1} = \frac{d_1 \times d_3}{d_2 \times d_4} \quad \text{or} \quad \frac{N_4}{150} = \frac{750 \times 900}{450 \times 150} = 10$$

$\therefore N_4 = 150 \times 10 = 1500$ r.p.m. **Ans.**

2. When there is a slip of 2% at each drive

We know that
$$\frac{N_4}{N_1} = \frac{d_1 \times d_3}{d_2 \times d_4} \left(1 - \frac{s_1}{100}\right) \left(1 - \frac{s_2}{100}\right)$$

$$\frac{N_4}{150} = \frac{750 \times 900}{450 \times 150} \left(1 - \frac{2}{100}\right) \left(1 - \frac{2}{100}\right) = 9.6$$

$\therefore N_4 = 150 \times 9.6 = 1440$ r.p.m. **Ans.**

Q.2 The power is transmitted from a pulley 1 m diameter running at 200 r.p.m. to a pulley 2.25 m diameter by means of a belt. Find the speed lost by the driven pulley as a result of creep, if the stress on the tight and slack side of the belt is 1.4 MPa and 0.5 MPa respectively. The Young's modulus for the material of the belt is 100 MPa.

Solution. Given : $d_1 = 1$ m ; $N_1 = 200$ r.p.m. ; $d_2 = 2.25$ m ; $\sigma_1 = 1.4$ MPa = 1.4×10^6 N/m² ; $\sigma_2 = 0.5$ MPa = 0.5×10^6 N/m² ; $E = 100$ MPa = 100×10^6 N/m²

Let N_2 = Speed of the driven pulley.

Neglecting creep, we know that

$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \quad \text{or} \quad N_2 = N_1 \times \frac{d_1}{d_2} = 200 \times \frac{1}{2.25} = 88.9 \text{ r.p.m.}$$

Considering creep, we know that

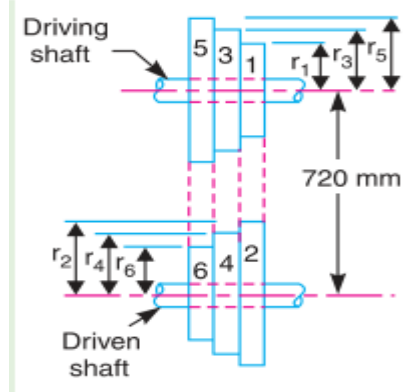
$$\frac{N_2}{N_1} = \frac{d_1}{d_2} \times \frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}$$

or
$$N_2 = 200 \times \frac{1}{2.25} \times \frac{100 \times 10^6 + \sqrt{0.5 \times 10^6}}{100 \times 10^6 + \sqrt{1.4 \times 10^6}} = 88.7 \text{ r.p.m.}$$

$\therefore$ Speed lost by driven pulley due to creep

$$= 88.9 - 88.7 = 0.2 \text{ r.p.m.} \quad \text{Ans.}$$

Q.3 A shaft which rotates at a constant speed of 160 r.p.m. is connected by belting to a parallel shaft 720 mm apart, which has to run at 60, 80 and 100 r.p.m. The smallest pulley on the driving shaft is 40 mm in radius. Determine the remaining radii of the two stepped pulleys for
1. a crossed belt, and 2. an open belt. Neglect belt thickness and slip.



Solution. Given : $N_1 = N_3 = N_5 = 160$ r.p.m. ; $x = 720$ mm ;
 $N_2 = 60$ r.p.m.; $N_4 = 80$ r.p.m.; $N_6 = 100$ r.p.m. ; $r_1 = 40$ mm

1. For a crossed belt

We know that for pulleys 1 and 2,

$$\frac{N_2}{N_1} = \frac{r_1}{r_2}$$

or
$$r_2 = r_1 \times \frac{N_1}{N_2} = 40 \times \frac{160}{60} = 106.7 \text{ mm Ans.}$$

and for pulleys 3 and 4,

$$\frac{N_4}{N_3} = \frac{r_3}{r_4} \text{ or } r_4 = r_3 \times \frac{N_3}{N_4} = r_3 \times \frac{160}{80} = 2r_3$$

We know that for a crossed belt drive,

$$r_1 + r_2 = r_3 + r_4 = r_5 + r_6 = 40 + 106.7 = 146.7 \text{ mm} \quad \dots(i)$$

$\therefore r_3 + 2r_3 = 146.7 \text{ or } r_3 = 146.7/3 = 48.9 \text{ mm Ans.}$

and $r_4 = 2r_3 = 2 \times 48.9 = 97.8 \text{ mm Ans.}$

Now for pulleys 5 and 6,

$$\frac{N_6}{N_5} = \frac{r_5}{r_6} \text{ or } r_6 = r_5 \times \frac{N_5}{N_6} = r_5 \times \frac{160}{100} = 1.6r_5$$

From equation (i),

$$r_5 + 1.6r_5 = 146.7 \text{ or } r_5 = 146.7/2.6 = 56.4 \text{ mm Ans.}$$

and $r_6 = 1.6r_5 = 1.6 \times 56.4 = 90.2 \text{ mm Ans.}$

2. For an open belt

We know that for pulleys 1 and 2,

$$\frac{N_2}{N_1} = \frac{r_1}{r_2} \text{ or } r_2 = r_1 \times \frac{N_1}{N_2} = 40 \times \frac{160}{60} = 106.7 \text{ mm Ans.}$$

and for pulleys 3 and 4,

$$\frac{N_4}{N_3} = \frac{r_3}{r_4} \text{ or } r_4 = r_3 \times \frac{N_3}{N_4} = r_3 \times \frac{160}{80} = 2r_3$$

We know that length of belt for an open belt drive,

$$\begin{aligned} L &= \pi(r_1 + r_2) + \frac{(r_2 - r_1)^2}{x} + 2x \\ &= \pi(40 + 106.7) + \frac{(106.7 - 40)^2}{720} + 2 \times 720 = 1907 \text{ mm} \end{aligned}$$

Since the length of the belt in an open belt drive is constant, therefore for pulleys 3 and 4, length of the belt (L),

$$1907 = \pi(r_3 + r_4) + \frac{(r_4 - r_3)^2}{x} + 2x$$

$$\begin{aligned}
 &= \pi(r_3 + 2r_3) + \frac{(2r_3 - r_3)^2}{720} + 2 \times 720 \\
 &= 9.426 r_3 + 0.0014 (r_3)^2 + 1440 \\
 \text{or } &0.0014 (r_3)^2 + 9.426 r_3 - 467 = 0 \\
 \therefore r_3 &= \frac{-9.426 \pm \sqrt{(9.426)^2 + 4 \times 0.0014 \times 467}}{2 \times 0.0014}
 \end{aligned}$$

$$= \frac{-9.426 \pm 9.564}{0.0028} = 49.3 \text{ mm Ans.}$$

and $r_4 = 2 r_3 = 2 \times 49.3 = 98.6 \text{ mm Ans.}$

Now for pulleys 5 and 6,

$$\begin{aligned}
 \frac{N_6}{N_5} &= \frac{r_5}{r_6} \quad \text{or} \\
 r_6 &= \frac{N_5}{N_6} \times r_5 = \frac{160}{100} \times r_5 = 1.6 r_5
 \end{aligned}$$

and length of the belt (L),

$$\begin{aligned}
 1907 &= \pi(r_5 + r_6) + \frac{(r_6 - r_5)^2}{x} + 2x \\
 &= \pi(r_5 + 1.6r_5) + \frac{(1.6r_5 - r_5)^2}{720} + 2 \times 720 \\
 &= 8.17 r_5 + 0.0005 (r_5)^2 + 1440
 \end{aligned}$$

or $0.0005 (r_5)^2 + 8.17 r_5 - 467 = 0$

$$\therefore r_5 = \frac{-8.17 \pm \sqrt{(8.17)^2 + 4 \times 0.0005 \times 467}}{2 \times 0.0005}$$

$$= \frac{-8.17 \pm 8.23}{0.001} = 60 \text{ mm Ans.}$$

...(Taking +ve sign)

and $r_6 = 1.6 r_5 = 1.6 \times 60 = 96 \text{ mm}$

Q.4 Find the power transmitted by a belt running over a pulley of 600 mm diameter at 200 r.p.m. The coefficient of friction between the belt and the pulley is 0.25, angle of lap 160° and maximum tension in the belt is 2500 N.

Solution. Given : $d = 600 \text{ mm} = 0.6 \text{ m}$; $N = 200 \text{ r.p.m.}$; $\mu = 0.25$; $\theta = 160^\circ = 160 \times \pi / 180 = 2.793 \text{ rad}$; $T_1 = 2500 \text{ N}$

We know that velocity of the belt,

$$v = \frac{\pi d \cdot N}{60} = \frac{\pi \times 0.6 \times 200}{60} = 6.284 \text{ m/s}$$

Let

$T_2 =$ Tension in the slack side of the belt.

We know that $2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta = 0.25 \times 2.793 = 0.6982$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.6982}{2.3} = 0.3036$$

$\therefore \frac{T_1}{T_2} = 2.01$... (Taking antilog of 0.3036)

$$T_2 = \frac{T_1}{2.01} = \frac{2500}{2.01} = 1244 \text{ N}$$

We know that power transmitted by the belt,

$$P = (T_1 - T_2) v = (2500 - 1244) 6.284 = 7890 \text{ W} \\ = 7.89 \text{ kW} \text{ Ans.}$$

Q.5 Two pulleys, one 450 mm diameter and the other 200 mm diameter are on parallel shafts 1.95 m apart and are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. What power can be transmitted by the belt when the larger pulley rotates at 200 rev/min, if the maximum permissible tension in the belt is 1 kN, and the coefficient of friction between the belt and pulley is 0.25 ?

Solution. Given : $d_1 = 450 \text{ mm} = 0.45 \text{ m}$ or $r_1 = 0.225 \text{ m}$; $d_2 = 200 \text{ mm} = 0.2 \text{ m}$ or $r_2 = 0.1 \text{ m}$; $x = 1.95 \text{ m}$; $N_1 = 200 \text{ r.p.m.}$; $T_1 = 1 \text{ kN} = 1000 \text{ N}$; $\mu = 0.25$

Solution. Given : $d_1 = 450 \text{ mm} = 0.45 \text{ m}$ or $r_1 = 0.225 \text{ m}$; $d_2 = 200 \text{ mm} = 0.2 \text{ m}$ or $r_2 = 0.1 \text{ m}$; $x = 1.95 \text{ m}$; $N_1 = 200 \text{ r.p.m.}$; $T_1 = 1 \text{ kN} = 1000 \text{ N}$; $\mu = 0.25$

We know that speed of the belt,

$$v = \frac{\pi d_1 \cdot N_1}{60} = \frac{\pi \times 0.45 \times 200}{60} = 4.714 \text{ m/s}$$

Length of the belt

We know that length of the crossed belt,

$$L = \pi(r_1 + r_2) + 2x + \frac{(r_1 + r_2)^2}{x} \\ = \pi(0.225 + 0.1) + 2 \times 1.95 + \frac{(0.225 + 0.1)^2}{1.95} = 4.975 \text{ m} \text{ Ans.}$$

Angle of contact between the belt and each pulley

Let

$\theta =$ Angle of contact between the belt and each pulley.

We know that for a crossed belt drive,

$$\sin \alpha = \frac{r_1 + r_2}{x} = \frac{0.225 + 0.1}{1.95} = 0.1667 \text{ or } \alpha = 9.6^\circ$$

$\therefore \theta = 180^\circ + 2\alpha = 180^\circ + 2 \times 9.6^\circ = 199.2^\circ$

$$= 199.2 \times \frac{\pi}{180} = 3.477 \text{ rad} \text{ Ans.}$$

Let T_2 = Tension in the slack side of the belt.

We know that

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta = 0.25 \times 3.477 = 0.8692$$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.8692}{2.3} = 0.378 \quad \text{or} \quad \frac{T_1}{T_2} = 2.387 \quad \dots (\text{Taking antilog of } 0.378)$$

$$\therefore T_2 = \frac{T_1}{2.387} = \frac{1000}{2.387} = 419 \text{ N}$$

We know that power transmitted,

$$P = (T_1 - T_2) v = (1000 - 419) 4.714 = 2740 \text{ W} = 2.74 \text{ kW} \quad \text{Ans.}$$

Q.6 A leather belt is required to transmit 7.5 kW from a pulley 1.2 m in diameter, running at 250 r.p.m. The angle embraced is 165° and the coefficient of friction between the belt and the pulley is 0.3. If the safe working stress for the leather belt is 1.5 MPa, density of leather 1 Mg/m³ and thickness of belt 10 mm, determine the width of the belt taking centrifugal tension into account.

Solution. Given : $P = 7.5 \text{ kW} = 7500 \text{ W}$; $d = 1.2 \text{ m}$; $N = 250 \text{ r.p.m.}$; $\theta = 165^\circ = 165 \times \pi / 180 = 2.88 \text{ rad}$; $\mu = 0.3$; $\sigma = 1.5 \text{ MPa} = 1.5 \times 10^6 \text{ N/m}^2$; $\rho = 1 \text{ Mg/m}^3 = 1 \times 10^6 \text{ g/m}^3 = 1000 \text{ kg/m}^3$; $t = 10 \text{ mm} = 0.01 \text{ m}$

Let b = Width of belt in metres, T_1 = Tension in the tight side of the belt in N, and T_2 = Tension in the slack side of the belt in N.

We know that velocity of the belt,

$$v = \pi d \cdot N / 60 = \pi \times 1.2 \times 250 / 60 = 15.71 \text{ m/s}$$

and power transmitted (P),

$$7500 = (T_1 - T_2) v = (T_1 - T_2) 15.71$$

$$\therefore T_1 - T_2 = 7500 / 15.71 = 477.4 \text{ N} \quad \dots (i)$$

We know that

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta = 0.3 \times 2.88 = 0.864$$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.864}{2.3} = 0.3756 \quad \text{or} \quad \frac{T_1}{T_2} = 2.375 \quad \dots (ii)$$

...(Taking antilog of 0.3756)

From equations (i) and (ii),

$$T_1 = 824.6 \text{ N, and } T_2 = 347.2 \text{ N}$$

We know that mass of the belt per metre length,

$$\begin{aligned} m &= \text{Area} \times \text{length} \times \text{density} = b \cdot t \cdot l \cdot \rho \\ &= b \times 0.01 \times 1 \times 1000 = 10 b \text{ kg} \end{aligned}$$

$\therefore$ Centrifugal tension,

$$T_C = m \cdot v^2 = 10 b (15.71)^2 = 2468 b \text{ N}$$

and maximum tension in the belt,

$$T = \sigma \cdot b \cdot t = 1.5 \times 10^6 \times b \times 0.01 = 15\,000 b \text{ N}$$

$$\text{We know that } T = T_1 + T_C \quad \text{or} \quad 15000 b = 824.6 + 2468 b$$

$$15\,000 b - 2468 b = 824.6 \quad \text{or} \quad 12\,532 b = 824.6$$

$$\therefore b = 824.6 / 12532 = 0.0658 \text{ m} = 65.8 \text{ mm} \quad \text{Ans.}$$

Gear drive

Gears are compact power transmission device that controls the speed, torque, and direction of rotation of the driven shaft. Gears may be classified into five main categories: Spur, Helical, Bevel, Hypoid, and Worm. Shaft orientation, efficiency, and speed determine the application of gear drive. Gears are a toothed disc, which transmits power from one shaft to another shaft by meshing with teeth of other gear.

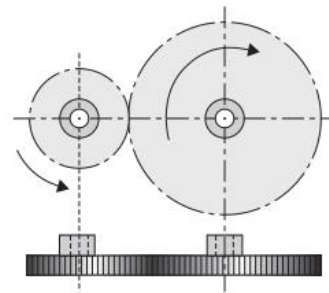
Classification of gears

(1) Gears can be classified according to the position of shafts as:

(1) Parallel Shafts

Spur Gears

General: Spur gears are the most commonly used gear. They are characterized by teeth which are parallel to the axis.



Advantages: Spur gears are easy to find, inexpensive, and efficient.

Limitations: Spur gears generally cannot be used when a direction change between the two shafts is required. Also, this type of gears is used for smaller speed due to noise creation at high-speed power transmission.

Helical Gears

Helical gears are similar to the spur gear except that the teeth are at an angle to the shaft, rather than parallel to its axis as in a spur gear. The longer teeth cause helical gears to have the following differences from spur gears of the same size:

(i) Tooth strength is greater because the teeth are longer.



(ii) Greater surface contact on the teeth allows a helical gear to carry more load than a spur gear.

(iii) The longer surface of contact reduces the efficiency of a helical gear relative to a spur gear.

Advantages:

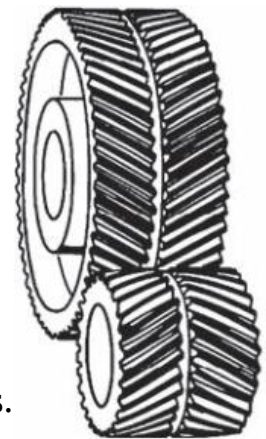
Helical gears can be used on non-parallel and even perpendicular shafts, and can carry higher loads than can spur gears.

Limitations:

Helical gears are expensive and much more difficult to manufacture. They are also slightly less efficient than a spur gear of the same size.

Double Helical / Herringbone Gears

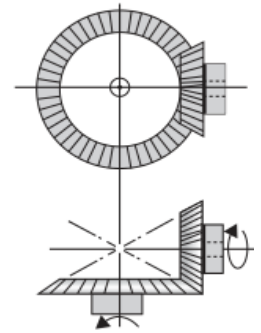
Double helical gears have one a right-hand helix and the other a left-hand helix. The teeth of two rows are separated by a groove used for tool run out. Axial thrust, which occurs in the case of single helical gears, two rows of teeth cancel each other. This can be run at high speeds with less noise and vibrations.



Intersecting Shaft

Bevel Gears

Bevel gears are primarily used to transfer power between intersecting shafts. The teeth of these gears are formed on a conical surface. Standard bevel gears have teeth



which are cut straight and are all parallel to the line pointing the apex of the cone on which the teeth are based. Spiral bevel gears are also available which teeth form arcs. Hypocycloid bevel gears are a special type of spiral gear that will allow non-intersecting, non-parallel shafts to mesh. Straight tool bevel gears are generally considered the best choice for systems with lower speeds. They become noisy above this point. One of the most common applications of bevel gears is the differential in automobiles.

Advantages:

It is an excellent choice for intersecting shaft systems.

Limitations:

It cannot be used for parallel shafts and becomes noisy at high speeds.

Hypoid Gears

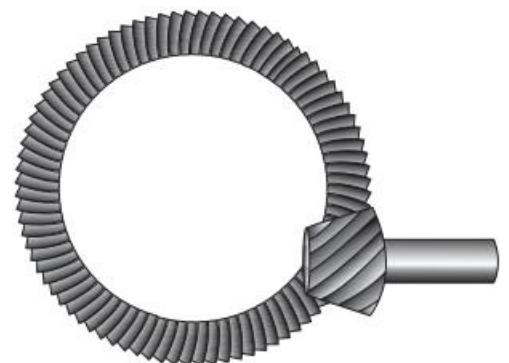
Hypoid gears resemble spiral bevels, but the axes of the pinion shaft and gear shaft do not intersect.

This configuration allows both shafts to be supported at both ends. In hypoid gears, the

meshing point of the pinion with the driven gear is about midway between the central position of a

pinion in a spiral-bevel and the extreme top or bottom position of

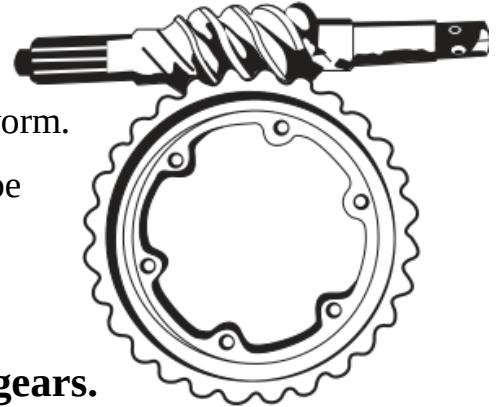
a worm. This geometry allows the driving and driven shafts to continue past each other so that end support bearings can be mounted. These bearings provide



greater rigidity than the support provided by the cantilever mounting used in some bevel gearing.

Worm Gears

If a tooth of a helical gear makes complete revolutions on the pitch cylinder, the resulting gear is known as a worm. The mating gear is called worm wheel. The worm may be a single start, double start, or triple start.



2. According to the peripheral velocity of the gears.

The gears, according to the peripheral velocity of the gears may be classified as : (a) Low velocity, (b) Medium velocity, and (c) High velocity.

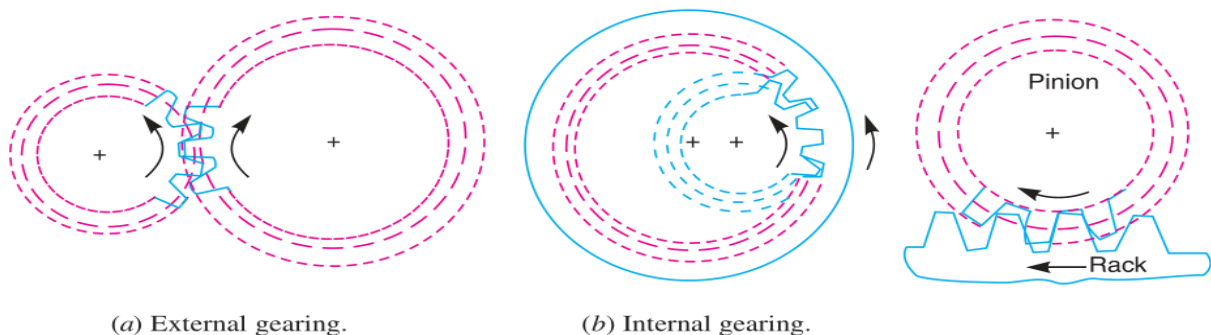
The gears having velocity less than 3 m/s are termed as low velocity gears and gears having velocity between 3 and 15 m/s are known as medium velocity gears. If the velocity of gears is more than 15 m/s, then these are called high speed gears

3. According to the type of gearing.

The gears, according to the type of gearing may be classified as : (a) External gearing, (b) Internal gearing, and (c) Rack and pinion.

In external gearing

The gears of the two shafts mesh externally with each other. The larger of these two wheels is called spur wheel and the smaller wheel is called pinion. In an external gearing, the motion of the two wheels is always unlike



(a) External gearing.

(b) Internal gearing.

In internal gearing

The gears of the two shafts mesh internally with each other. The larger of these two wheels is called annular wheel and the smaller wheel is called pinion. In an internal gearing, the motion of the two wheels is always like,

Rack and pinion

Sometimes, the gear of a shaft meshes externally and internally with the gears in a *straight line, Such type of gear is called rack and pinion. The straight line gear is called rack and the circular wheel is called pinion. A little consideration will show that with the help of a rack and pinion, we can convert linear motion into rotary motion

4. According to position of teeth on the gear surface.

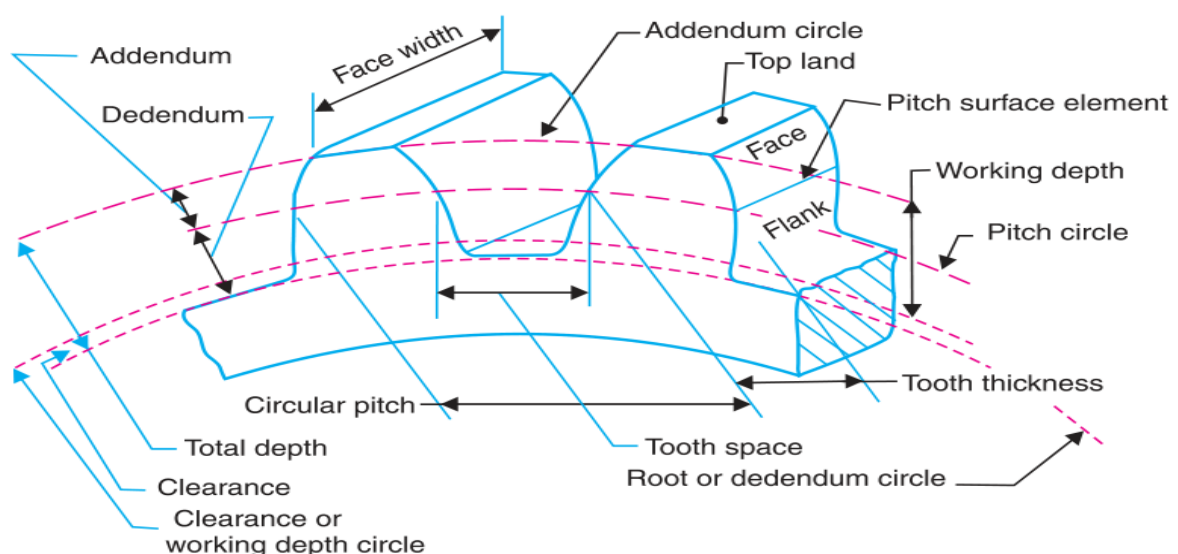
The teeth on the gear surface may be

(a) straight, (b) inclined, and (c) curved.

We have discussed earlier that the spur gears have straight teeth whereas helical gears have their teeth inclined to the wheel rim. In case of spiral gears, the teeth are curved over the rim surface.



Terms Used in Gears



(1). Pitch circle diameter. It is the diameter of the pitch circle. The size of the gear is usually specified by the pitch circle diameter. It is also known as pitch diameter.

(2). Pressure angle or angle of obliquity. It is the angle between the common normal to two gear teeth at the point of contact and the common tangent at the pitch point. It is usually denoted by ϕ . The standard pressure angles are 14.5° and 20° .

(3). Addendum. It is the radial distance of a tooth from the pitch circle to the top of the tooth.

(4). Dedendum. It is the radial distance of a tooth from the pitch circle to the bottom of the tooth.

(5). Circular pitch. It is the distance measured on the circumference of the pitch circle from a point of one tooth to the corresponding point on the next tooth. It is usually denoted by p_c .

Mathematically, Circular pitch, $p_c = \pi D/T$

Where D = Diameter of the pitch circle, and

T = Number of teeth on the wheel.

A little consideration will show that the two gears will mesh together correctly, if the two wheels have the same circular pitch.

If D_1 and D_2 are the diameters of the two meshing gears having the teeth T_1 and T_2 respectively, then for them to mesh correctly,

$$p_c = \frac{\pi D_1}{T_1} = \frac{\pi D_2}{T_2} \quad \text{or} \quad \frac{D_1}{D_2} = \frac{T_1}{T_2}$$

(6). Diametral pitch. It is the ratio of number of teeth to the pitch circle diameter in millimetres. It is denoted by p_d . Mathematically,

Diametral pitch, $p_d = \frac{T}{D} = \frac{\pi}{p_c}$... $\left(\because p_c = \frac{\pi D}{T} \right)$

where T = Number of teeth, and
 D = Pitch circle diameter.

(7). Module. It is the ratio of the pitch circle diameter in millimeters to the number of teeth. It is usually denoted by m . Mathematically,

$$\text{Module, } m = D/T$$

Gear Materials

The material used for the manufacture of gears depends upon the strength and service conditions like wear, noise etc. The gears may be manufactured from metallic or non-metallic materials.

The metallic gears with cut teeth are commercially obtainable in cast iron, steel and bronze. **The nonmetallic materials** like wood, raw hide, compressed paper and synthetic resins like nylon are used for gears, especially for reducing noise.

The cast iron is widely used for the manufacture of gears due to its good wearing properties, excellent machinability and ease of producing complicated shapes by casting method. The cast iron gears with cut teeth may be employed, where smooth action is not important.

The steel is used for high strength gears and steel may be plain carbon steel or alloy steel. The steel gears are usually heat treated in order to combine properly the toughness and tooth hardness.

The phosphor bronze is widely used for worm gears in order to reduce wear of the worms which will be excessive with cast iron or steel.

Gear

Gear Trains

Sometimes, two or more gears are made to mesh with each other to transmit power from one shaft to another. Such a combination is called gear train or train of toothed wheels. The nature of the train used depends upon the velocity ratio required and the relative position of the axes of shafts. A gear train may consist of spur, bevel or spiral gears.

1. Simple gear train,
2. Compound gear train,
3. Reverted gear train,
4. Epicyclic gear train.

Simple Gear Train

When there is only one gear on each shaft, it is known as simple gear train. The gears are represented by their pitch circles. When the distance between the two shafts is small, the two gears 1 and 2 are made to mesh with each other to transmit motion from one shaft to the other. Since the gear 1 drives the gear 2, therefore gear 1 is called the driver and the gear 2 is called the driven or follower. It may be noted that the motion of the driven gear is opposite to the motion of driving gear.

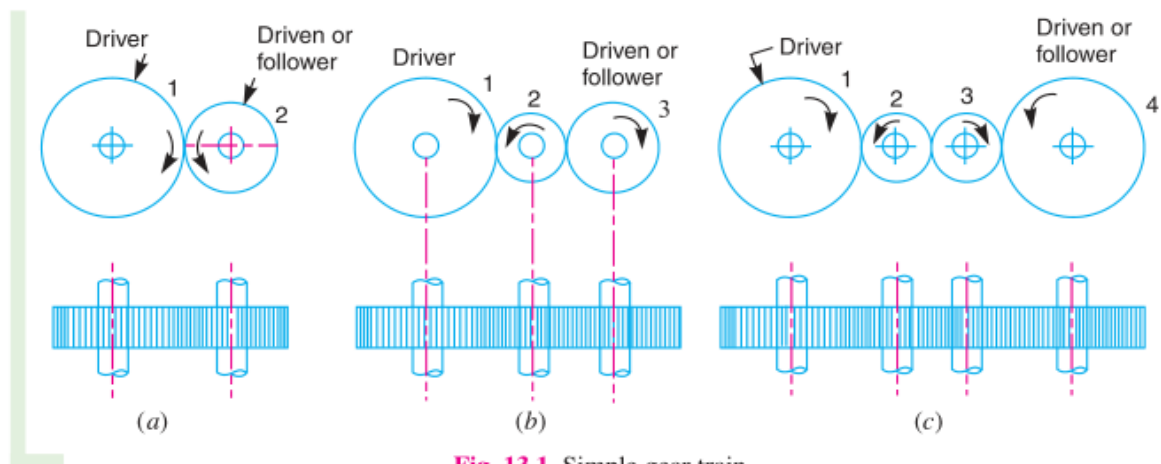


Fig. 13.1. Simple gear train.

Let

N_1 = Speed of gear 1 (or driver) in r.p.m.,

N_2 = Speed of gear 2 (or driven or follower) in r.p.m.,

T_1 = Number of teeth on gear 1, and

T_2 = Number of teeth on gear 2.

Since the speed ratio (or velocity ratio) of gear train is the ratio of the speed of the driver to the speed of the driven or follower and ratio of speeds of any pair of gears in mesh is the inverse of their number of teeth, therefore

$$\text{Speed ratio} = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

It may be noted that ratio of the speed of the driven or follower to the speed of the driver is known as **train value** of the gear train. Mathematically,

$$\text{Train value} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$

N_3 = Speed of driven or follower in r.p.m.,
 T_1 = Number of teeth on driver,
 T_2 = Number of teeth on intermediate gear, and
 T_3 = Number of teeth on driven or follower.

Since the driving gear 1 is in mesh with the intermediate gear 2, therefore speed ratio for these two gears is

$$\frac{N_1}{N_2} = \frac{T_2}{T_1} \quad \dots(i)$$

Similarly, as the intermediate gear 2 is in mesh with the driven gear 3, therefore speed ratio for these two gears is

$$\frac{N_2}{N_3} = \frac{T_3}{T_2} \quad \dots(ii)$$

The speed ratio of the gear train as shown in Fig. 13.1 (b) is obtained by multiplying the equations (i) and (ii).

$$\therefore \frac{N_1}{N_2} \times \frac{N_2}{N_3} = \frac{T_2}{T_1} \times \frac{T_3}{T_2} \quad \text{or} \quad \frac{N_1}{N_3} = \frac{T_3}{T_1}$$

i.e.

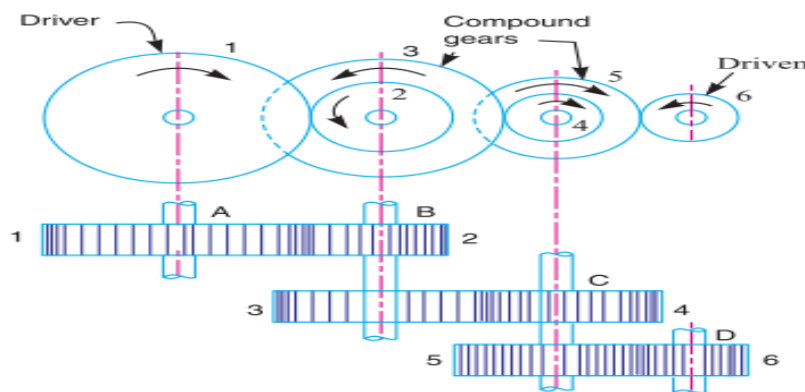
$$\text{Speed ratio} = \frac{\text{Speed of driver}}{\text{Speed of driven}} = \frac{\text{No. of teeth on driven}}{\text{No. of teeth on driver}}$$

and

$$\text{Train value} = \frac{\text{Speed of driven}}{\text{Speed of driver}} = \frac{\text{No. of teeth on driver}}{\text{No. of teeth on driven}}$$

Compound Gear Train

When there are more than one gear on a shaft, it is called a compound train of gear. The idle gears, in a simple train of gears do not effect the speed ratio of the system. But these gears are useful in bridging over the space between the driver and the drive But whenever the distance between the driver and the driven or follower has to be bridged over by intermediate gears and at the same time a great (or much less) speed ratio is required, then the advantage of intermediate gears is intensified by providing compound gears on intermediate shafts



Let

N_1 = Speed of driving gear 1,

T_1 = Number of teeth on driving gear 1,

$N_2, N_3, \dots, N_6$ = Speed of respective gears in r.p.m., and

$T_2, T_3, \dots, T_6$ = Number of teeth on respective gears.

Since gear 1 is in mesh with gear 2, therefore its speed ratio is

$$\frac{N_1}{N_2} = \frac{T_2}{T_1} \quad \dots(i)$$

Similarly, for gears 3 and 4, speed ratio is

$$\frac{N_3}{N_4} = \frac{T_4}{T_3} \quad \dots(ii)$$

and for gears 5 and 6, speed ratio is

$$\frac{N_5}{N_6} = \frac{T_6}{T_5} \quad \dots(iii)$$

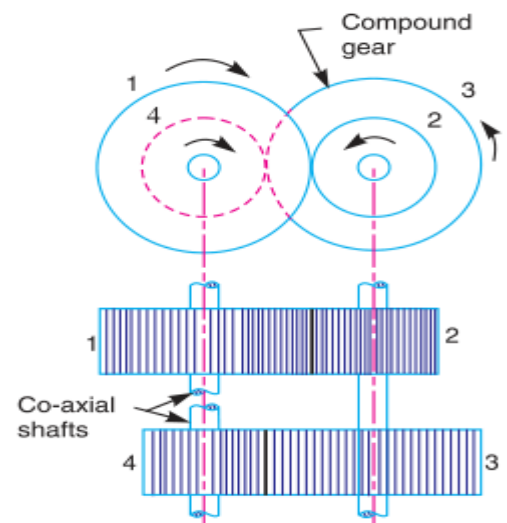
The speed ratio of compound gear train is obtained by multiplying the equations (i), (ii) and (iii),

$$\therefore \frac{N_1}{N_2} \times \frac{N_3}{N_4} \times \frac{N_5}{N_6} = \frac{T_2}{T_1} \times \frac{T_4}{T_3} \times \frac{T_6}{T_5} \quad \text{or} \quad \frac{N_1}{N_6} = \frac{T_2 \times T_4 \times T_6}{T_1 \times T_3 \times T_5}$$

* Since gears 2 and 3 are mounted on one shaft B, therefore $N_2 = N_3$. Similarly gears 4 and 5 are mounted on shaft C, therefore $N_4 = N_5$.

Reverted Gear Train

When the axes of the first gear (i.e. first driver) and the last gear (i.e. last driven or follower) are co-axial, then the gear train is known as reverted gear train. That gear 1 (i.e. first driver) drives the gear 2 (i.e. first driven or follower) in the opposite direction. Since the gears 2 and 3 are mounted on the same shaft, therefore they form a compound gear and the gear 3 will rotate in the same direction as that of gear 2. The gear 3 (which is now the second driver) drives the gear 4 (i.e. the last driven or follower) in the same direction as that of gear 1. Thus we see that in a reverted gear train, the motion of the first gear and the last gear is like.



Let T_1 = Number of teeth on gear 1,
 r_1 = Pitch circle radius of gear 1, and
 N_1 = Speed of gear 1 in r.p.m.

Similarly,

T_2, T_3, T_4 = Number of teeth on respective gears,
 r_2, r_3, r_4 = Pitch circle radii of respective gears, and
 N_2, N_3, N_4 = Speed of respective gears in r.p.m.

Since the distance between the centres of the shafts of gears 1 and 2 as well as gears 3 and 4 is same, therefore

$$r_1 + r_2 = r_3 + r_4 \quad \dots(i)$$

Also, the circular pitch or module of all the gears is assumed to be same, therefore number of teeth on each gear is directly proportional to its circumference or radius.

$$\therefore T_1 + T_2 = T_3 + T_4 \quad \dots(ii)$$

and

$$\text{Speed ratio} = \frac{\text{Product of number of teeth on drivers}}{\text{Product of number of teeth on driven}}$$

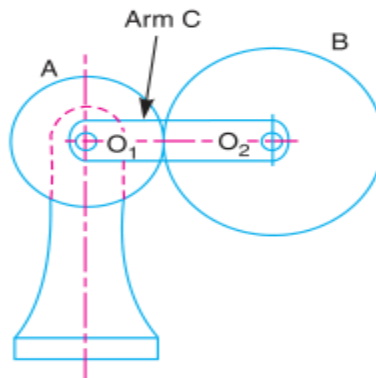
or

$$\frac{N_1}{N_4} = \frac{T_2 \times T_4}{T_1 \times T_3} \quad \dots(iii)$$

Epicyclic Gear Train

In an epicyclic gear train, the axes of the shafts, over which the gears are mounted, may move relative to a fixed axis. A simple epicyclic gear train where a gear A and the arm C have a common axis at O_1 about which they can rotate. The gear B meshes with gear A and has its axis on the arm at O_2 , about which the gear B can rotate. If the arm is fixed, the gear train is simple and gear A can drive gear B or vice-versa, but if gear A is fixed and the arm is rotated about the axis of gear A (i.e. O_1), then the gear B is forced to rotate upon and around gear A. Such a motion is called epicyclic and the gear trains arranged in such a manner that one or more of their members move upon and around another member are known as epicyclic gear trains (epi. means upon and cyclic means around). The epicyclic gear trains may be simple or compound. The epicyclic gear trains are useful for transmitting high velocity ratios with gears of moderate size in a comparatively lesser space. The epicyclic gear trains are used

in the back gear of lathe, differential gears of the automobiles, hoists, pulley blocks, wrist watches etc



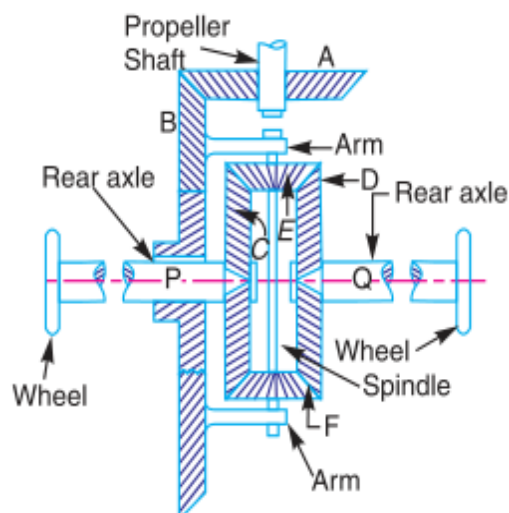
Velocity Ratio of Epicyclic Gear Train

Tabular method,

Let T_A = Number of teeth on gear A, and T_B = Number of teeth on gear B.

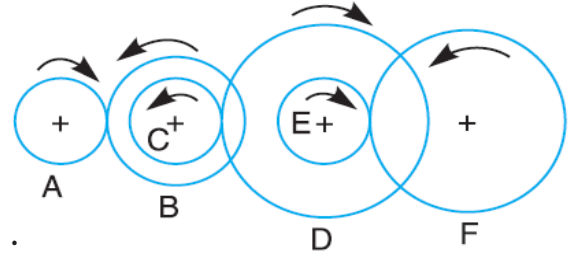
Step No.	Conditions of motion	Revolutions of elements		
		Arm C	Gear A	Gear B
1.	Arm fixed-gear A rotates through + 1 revolution <i>i.e.</i> 1 rev. anticlockwise	0	+ 1	$-\frac{T_A}{T_B}$
2.	Arm fixed-gear A rotates through + x revolutions	0	+ x	$-x \times \frac{T_A}{T_B}$
3.	Add + y revolutions to all elements	+ y	+ y	+ y
4.	Total motion	+ y	+ $x + y$	$y - x \times \frac{T_A}{T_B}$

Differential gear of an Automobile



(1) The gearing of a machine tool is shown in which, The motor shaft is connected to gear A and rotates at 975 r.p.m. The gear wheels B, C, D and E are fixed to parallel shafts rotating together. The final gear F is fixed on the output shaft. What is the speed of gear F? The number of teeth on each gear are as given below

Gear	A	B	C	D	E	F
No. of teeth	20	50	25	75	26	65



Solution. Given : $N_A = 975$ r.p.m. ;

$T_A = 20$; $T_B = 50$; $T_C = 25$; $T_D = 75$; $T_E = 26$;

$T_F = 65$

$$\frac{\text{Speed of the first driver}}{\text{Speed of the last driven}} = \frac{\text{Product of no. of teeth on drivers}}{\text{Product of no. of teeth on driven}}$$

$$\frac{N_A}{N_F} = \frac{T_B \times T_D \times T_F}{T_A \times T_C \times T_E} = \frac{50 \times 75 \times 65}{20 \times 25 \times 26} = 18.75$$

$$N_F = \frac{N_A}{18.75} = \frac{975}{18.75} = 52 \text{ r. p. m. } \textbf{Ans.}$$

(2) Two parallel shafts, about 600 mm apart are to be connected by spur gears. One shaft is to run at 360 r.p.m. and the other at 120 r.p.m. Calculate the Number of teeth on Spur gears, if the circular pitch is to be 25 mm.

Solution. Given : $x = 600$ mm ; $N_1 = 360$ r.p.m. ; $N_2 = 120$ r.p.m. ; $p_c = 25$ mm

Let

d_1 = Pitch circle diameter of the first gear, and

d_2 = Pitch circle diameter of the second gear.

We know that speed ratio,

$$\frac{N_1}{N_2} = \frac{d_2}{d_1} = \frac{360}{120} = 3 \quad \text{or} \quad d_2 = 3d_1 \quad \dots(i)$$

and centre distance between the shafts (x),

$$600 = \frac{1}{2} (d_1 + d_2) \quad \text{or} \quad d_1 + d_2 = 1200 \quad \dots(ii)$$

From equations (i) and (ii), we find that

$$d_1 = 300 \text{ mm, and } d_2 = 900 \text{ mm}$$

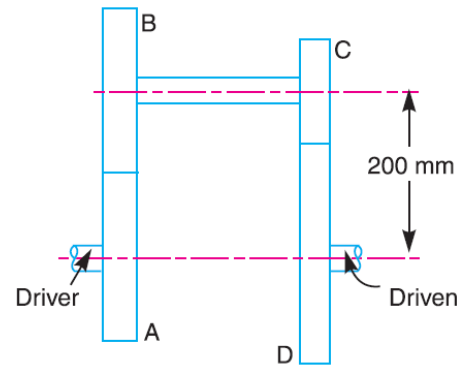
$\therefore$ Number of teeth on the first gear,

$$T_1 = \frac{\pi d_1}{p_c} = \frac{\pi \times 300}{25} = 37.7$$

and number of teeth on the second gear,

$$T_2 = \frac{\pi d_2}{p_c} = \frac{\pi \times 900}{25} = 113.1$$

(3) The speed ratio of the reverted gear train, is to be 12. The module pitch of gears A and B is 3.125 mm and of gears C and D is 2.5 mm. Calculate the suitable numbers of teeth for the gears. No gear is to have less than 24 teeth.



Solution. Given : Speed ratio, $N_A/N_D = 12$;
 $m_A = m_B = 3.125$ mm ; $m_C = m_D = 2.5$ mm

Let N_A = Speed of gear A,

T_A = Number of teeth on gear A,

r_A = Pitch circle radius of gear A,

N_B, N_C, N_D = Speed of respective gears,

T_B, T_C, T_D = Number of teeth on respective gears,

r_B, r_C, r_D = Pitch circle radii of respective gears.

Since the speed ratio between the gears A and B and between the gears C and D are to be same, therefore

$$* \frac{N_A}{N_B} = \frac{N_C}{N_D} = \sqrt{12} = 3.464$$

Also the speed ratio of any pair of gears in mesh is the inverse of their number of teeth, therefore

$$\frac{T_B}{T_A} = \frac{T_D}{T_C} = 3.464 \quad \dots(i)$$

We know that the distance between the shafts

$$x = r_A + r_B = r_C + r_D = 200 \text{ mm}$$

$$\text{or} \quad \frac{m_A \cdot T_A}{2} + \frac{m_B \cdot T_B}{2} = \frac{m_C \cdot T_C}{2} + \frac{m_D \cdot T_D}{2} = 200 \quad \dots \left(\because r = \frac{m \cdot T}{2} \right)$$

$$3.125 (T_A + T_B) = 2.5 (T_C + T_D) = 400 \quad \dots(\because m_A = m_B, \text{ and } m_C = m_D)$$

$$\therefore T_A + T_B = 400 / 3.125 = 128 \quad \dots(ii)$$

$$\text{and} \quad T_C + T_D = 400 / 2.5 = 160 \quad \dots(iii)$$

From equation (i), $T_B = 3.464 T_A$. Substituting this value of T_B in equation (ii),

$$T_A + 3.464 T_A = 128 \quad \text{or} \quad T_A = 128 / 4.464 = 28.67 \text{ say } 28 \text{ Ans.}$$

$$\text{and} \quad T_B = 128 - 28 = 100 \text{ Ans.}$$

Again from equation (i), $T_D = 3.464 T_C$. Substituting this value of T_D in equation (iii),

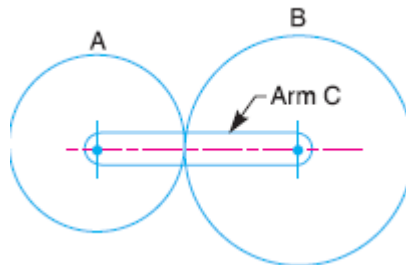
$$T_C + 3.464 T_C = 160 \quad \text{or} \quad T_C = 160 / 4.464 = 35.84 \text{ say } 36 \text{ Ans.}$$

$$\text{and} \quad T_D = 160 - 36 = 124 \text{ Ans.}$$

Note : The speed ratio of the reverted gear train with the calculated values of number of teeth on each gear is

$$\frac{N_A}{N_D} = \frac{T_B \times T_D}{T_A \times T_C} = \frac{100 \times 124}{28 \times 36} = 12.3$$

(4) In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 r.p.m. in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 r.p.m. in the clockwise direction, what will be the speed of gear B ?



Step No.	Conditions of motion	Revolutions of elements		
		Arm C	Gear A	Gear B
1.	Arm fixed-gear A rotates through + 1 revolution (i.e. 1 rev. anticlockwise)	0	+ 1	$-\frac{T_A}{T_B}$
2.	Arm fixed-gear A rotates through + x revolutions	0	+ x	$-x \times \frac{T_A}{T_B}$
3.	Add + y revolutions to all elements	+ y	+ y	+ y
4.	Total motion	+ y	x + y	$y - x \times \frac{T_A}{T_B}$

Speed of gear B when gear A is fixed

Since the speed of arm is 150 r.p.m. anticlockwise, therefore from the fourth row of the table,

$$y = + 150 \text{ r.p.m.}$$

Also the gear A is fixed, therefore

$$x + y = 0 \quad \text{or} \quad x = -y = - 150 \text{ r.p.m.}$$

$$\therefore \text{Speed of gear B, } N_B = y - x \times \frac{T_A}{T_B} = 150 + 150 \times \frac{36}{45} = + 270 \text{ r.p.m.}$$

$$= 270 \text{ r.p.m. (anticlockwise) \quad Ans.}$$

Speed of gear B when gear A makes 300 r.p.m. clockwise

Since the gear A makes 300 r.p.m. clockwise, therefore from the fourth row of the table,

$$x + y = - 300 \quad \text{or} \quad x = - 300 - y = - 300 - 150 = - 450 \text{ r.p.m.}$$

$\therefore$ Speed of gear B,

$$N_B = y - x \times \frac{T_A}{T_B} = 150 + 450 \times \frac{36}{45} = + 510 \text{ r.p.m.}$$

$$= 510 \text{ r.p.m. (anticlockwise) \quad Ans.}$$